

Renormalization-Group Flow in Three Dimensions

Source: *Perimeter Scholars International, Quantum Field Theory II, Tutorial 6; 7 worked parts.*

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1 Renormalization

Question prompts paraphrased from the official Perimeter PSI Online sheet; the equations and notation follow the source.

1.1 RG Flow in Three Dimensions

Consider massless Euclidean ϕ^4 theory in three spacetime dimensions,

$$S_E[\phi] = \int d^3x \left[\frac{1}{2}(\partial\phi)^2 + \frac{g}{4!}\phi^4 \right]. \quad (1.1)$$

1. Determine the mass dimension of the coupling g .

Solution. g has mass dimension $4 - d = 1$.

For two-to-two scattering with incoming momenta p_1, p_2 and outgoing momenta p_3, p_4 , define the Mandelstam variables

$$s = (p_1 + p_2)^2, \quad t = (p_1 - p_3)^2, \quad u = (p_1 - p_4)^2. \quad (1.2)$$

In the conventions of the sheet,

$$s + t + u = -4m^2 = 0, \quad (1.3)$$

where the final equality uses $m = 0$.

2. Write the four-point irreducible Green function $\bar{\Gamma}_{(4)}$ through one loop in terms of p_1, p_2, p_3, p_4 . Set up the loop integrals, but do not evaluate them yet.

Solution.

$$-\bar{\Gamma}_{(4)} = -g + \frac{(-g)^2}{2} \int d^d k \frac{1}{k^2(k+p_1+p_2)^2} + (p_2 \leftrightarrow -p_3) + (p_2 \leftrightarrow -p_4) + \dots \quad (1.4)$$

3. In four dimensions the corresponding integrals are ultraviolet divergent. Show, by power counting or direct analysis, that the three-dimensional loop integrals are ultraviolet finite. No UV regulator is needed here.

Solution. For $d = 4$, the integral is logarithmic divergent. For $d = 3$, it remains finite.

4. Show that

$$\bar{\Gamma}_{(4)}(s, t, u) = g - \frac{g^2 C}{2} \left[\sqrt{s^{-1}} + \sqrt{t^{-1}} + \sqrt{u^{-1}} \right], \quad C > 0. \quad (1.5)$$

Determining the precise value of C using Feynman parameters is optional.

Solution.

$$\begin{aligned} \bar{\Gamma}_{(4)} &= g - \frac{g^2}{2} \left(2\pi \int_{-1}^1 d\cos\theta \int_0^\infty dk \frac{1}{k^2 + s + 2\sqrt{s}\cos\theta} + (s \leftrightarrow t) + (s \leftrightarrow u) \right) \\ &= g - \frac{g^2}{2} \left(\frac{\pi}{\sqrt{s}} \int_0^\infty \frac{dk}{k} \log \left(\frac{k^2 + s + 2\sqrt{s}}{k^2 + s - 2\sqrt{s}} \right) + (s \leftrightarrow t) + (s \leftrightarrow u) \right) \\ &= g - \frac{g^2}{2} \left(\frac{2\pi}{\sqrt{s}} \int_0^\infty \frac{dk}{k} \log \left(\frac{k + \sqrt{s}}{k - \sqrt{s}} \right) + (s \leftrightarrow t) + (s \leftrightarrow u) \right) \\ &= g - \frac{g^2}{2} \left(\frac{2\pi}{\sqrt{s}} \int_0^\infty \frac{dk}{k} \log \left(\frac{k+1}{k-1} \right) + (s \leftrightarrow t) + (s \leftrightarrow u) \right) \\ &= g - \frac{g^2}{2} 2\pi \int_0^\infty \frac{dk}{k} \log \left(\frac{k+1}{k-1} \right) (\sqrt{s^{-1}} + \sqrt{t^{-1}} + \sqrt{u^{-1}}), \end{aligned} \quad (1.6)$$

where we have rescaled $k \rightarrow k/\sqrt{s}$ in the second last equality.

Although the symmetric choice $s = t = u = \mu^2$ is incompatible with the preceding Mandelstam relation, adopt it as the simplified subtraction point used in the sheet and define

$$h_R = \mu^{-1} \bar{\Gamma}_{(4)}(s = t = u = \mu^2). \quad (1.7)$$

Hence,

$$h_R = \mu^{-1} \left[g - \frac{3}{2} g^2 \mu^{-1} C \right]. \quad (1.8)$$

5. At an arbitrary energy E , define

$$h_{\text{eff}}(E) = E^{-1} \bar{\Gamma}_{(4)}(s = t = u = E^2), \quad (1.9)$$

together with

$$\beta(h_{\text{eff}}) = E \frac{dh_{\text{eff}}}{dE}. \quad (1.10)$$

Show, through order g^2 , that

$$\beta(h) = -h + \frac{3}{2}Ch^2 + \text{higher-order terms.} \quad (1.11)$$

Solution.

$$\beta(h) = E \left(-\frac{1}{E^2} \left(g - \frac{3}{2}g^2 E^{-1}C \right) + \frac{1}{E} \cdot \frac{3}{2}g^2 \frac{1}{E^2}C \right) = -h + \frac{3}{2}Ch^2 + \mathcal{O}(g^3). \quad (1.12)$$

6. Plot $\beta(h)$ as a function of h .

7. Suppose h is small and positive at some finite energy. Show that $h \rightarrow 0$ as $E \rightarrow \infty$, and identify this zero as the ultraviolet fixed point of the RG flow.

Solution. Solving the RG equation, we find

$$h(E) = \frac{1}{\frac{3}{2}C - \frac{h_0}{\frac{3}{2}Ch_0 - 1} \frac{E}{E_0}}. \quad (1.13)$$

Clearly, as $E \rightarrow \infty$, $h \rightarrow 0$. Note that $\beta(0) = 0$, and hence this is a UV fixed point.

8. Starting again from small positive h , show that the flow approaches a finite value h_0 as $E \rightarrow 0$, where h_0 is the nonzero root of $\beta(h)$. Interpret h_0 as an infrared fixed point.

Solution. As $E \rightarrow 0$, $h \rightarrow \frac{2}{3C}$. Note that $\beta(\frac{2}{3C}) = 0$, and hence this is an IR fixed point. This suggests the existence of the Wilson-Fisher fixed point. Note that at this point, the theory become strongly coupled, since all higher order terms become relevant.